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The Rebundling Hypothesis

What LLMs unbundle, and what active inference could restore

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github.com/egolabs-ai/iwai26

abstract · poster · code

1 THE DIAGNOSIS

Competence and understanding come apart. The LLM is a *resolution device*: what we treated as one capacity resolves into several.



Brooks's hexapod ***Genghis*** crossed rough terrain on locally coupled behaviours – no world model, no planner.

LLMs repeat the dissociation in language: what **Floridi** calls **agency without intelligence**.

BIOLOGY — LLM

Tomasello's four levels of agency, each added under selection



LEVEL 4 — **Normative** — humans · child >3 yr — in language



LEVEL 3 — **Rational** — great apes · preschooler — in language



LEVEL 2 — **Intentional** — early mammals · toddler



LEVEL 1 — **Goal-directed** — reptiles · infant <9 mo

… and recapitulated, in the same order, during development

NEVER BUILT

the agent is *inverted*: a top with no bottom

CLAIM 1

The **training objective** determines the bundle.

phylogenetic selection → training

ontogenetic life → deployment

Optimise survival and you get biology's bundle. Optimise next-token prediction and you get one **with no biological precedent**.

2 NINE DISENTANGLEMENTS

Pairs that biology fuses and the LLM separates. Under the free energy principle each fusion is *constitutive*, not contingent.

	LLM *has*	LLM *lacks*
MISSING BOTTOM · LEVELS 1–2		
	Agency (causal efficacy)	Intelligence (understanding)
	Feedback control	Existential grounding
	Modelling intentionality	Perceiving other minds
DISEMBODIED TOP · LEVELS 3–4		
	Reason-discourse	Reasoning
	Normative competence	Normative inhabitation
CROSS-CUTTING		
	Flexible behaviour	Emotional tendencies
	Phylogenetic learning (= training)	Ontogenetic learning (= development)
	Linguistic generativity	Behavioural generativity
	Statistical extraction (of regularities)	Core-knowledge priors (innate)

the cut

CLAIM 2

Rebundling is an **embodied** problem, not a linguistic one.

The LLM probes only the verbal face of this topology, so no amount of scale or text reaches the embodied bottom.

What LLMs lack is not access or reportability, but

sentience

– the felt stakes of a system with **something to lose**.

3 THE HYPOTHESIS

An agent trained and deployed under a **free-energy objective** with **homeostatic priors** and **embodied sensorimotor coupling** – the objective selection approximated – should recover the bundle, through the *dynamics that bind its components*, not the addition of features.

Homeostasis — a prior over preferred observations

Active inference — perception and action minimise free energy

Embodied touch — the self–world boundary, sensed from inside

bound under ONE free-energy objective

a self-evidencing agent

its persistence *is* the minimisation

4 THE TEST

Drop one component and a distinct signature of agency is predicted to go with it.



	homeostasis	active inference	embodied touch	predicted behavioural signature
Full bundle	●	●	●	self-other discrimination; proactive damage-avoidance; epistemic self-touch
no homeostasis	○	●	●	feels its body, but nothing is at stake
no active inference	●	○	●	acts only after harm – no anticipatory care
no embodied touch	●	●	○	no self–world boundary: self and world confused

The pairwise combinations are already studied; the claim concerns their **integration under one objective**.

THE PROGRAMME

1 Does a *simulated* homeostatic objective suffice, or must the substrate materially maintain itself?

2 Diagnostic work *scores* systems on indicators; rebundling **builds the substrate** they would measure.

3 The constitutive reading of the FEP is contested, so build such agents and test.

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